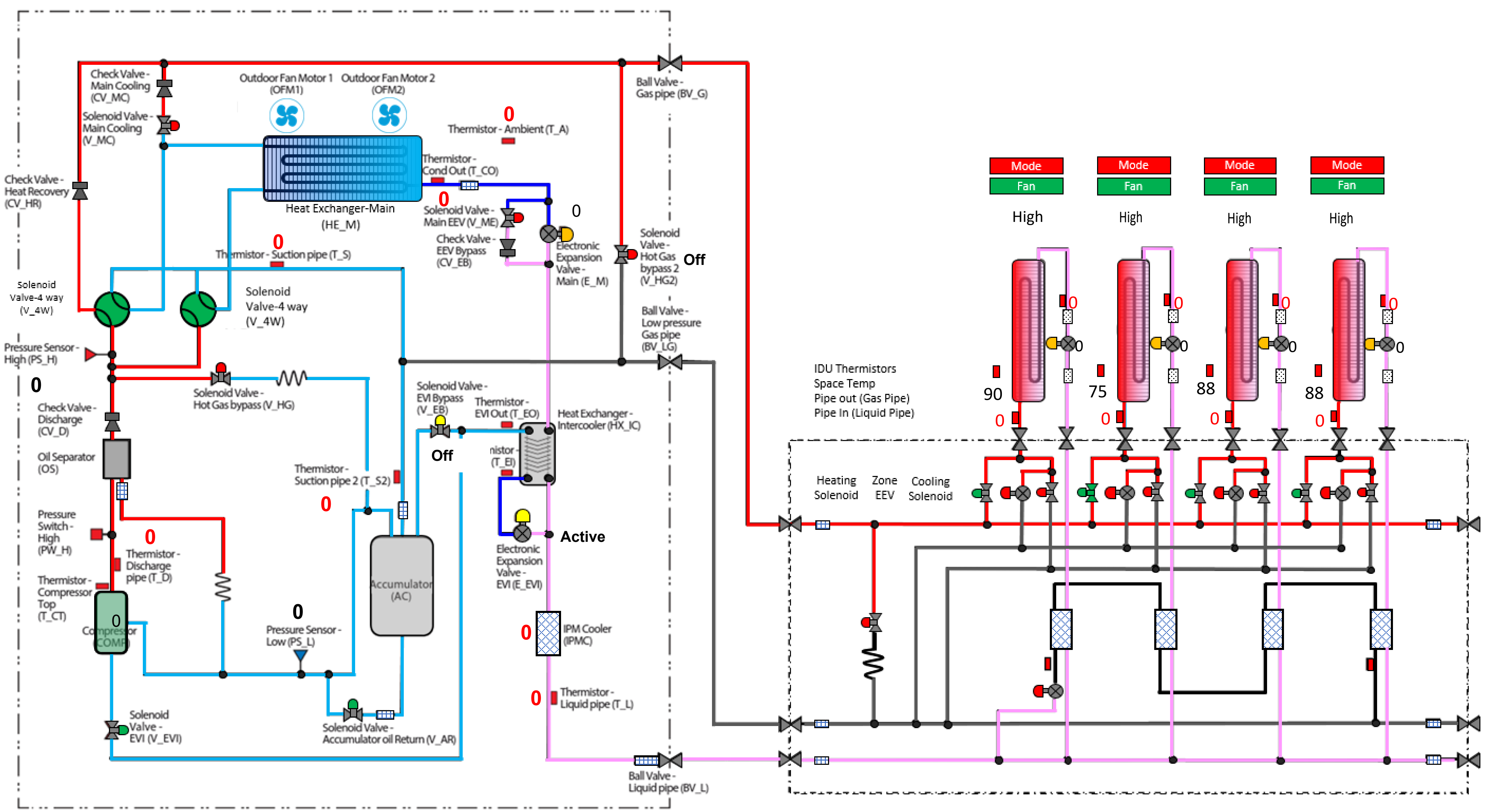
SAMSUNG

DVM S Heat Recovery Operation Evaluation
Heating Mode



Comments:

Outdoor Unit			
Component	**Target**	**Reading**	
Ambient Temp.	20-160	0	
Compressor Hz.		0	Ok
High pressure	427 psig	0	Ok
H.P Sat. Temp	121.5^0		
Discharge Temp.	150^0-212^0	0	Ok
IPM Temp,	Below 194^0	0	Ok
Low pressure		0	Ok
L.P. Sat. Temp	OAT - 2^0 - 10^0		
Suction 1 Temp.		0	Ok
Suction 2 Temp		0	Ok
S.H.	3.6^0 or 9^0	*0*	Ok
Main EEV Position		0	Ok
Cond. Out Temp.	OAT - 2^0	0	Ok
Liquid Line Temp.		0	Ok
Sub-Cooling @ T_L	9^0 +	*0*	Ok

Indoor Units			
Component	**Target**	**Reading**	
IDU 1 EEV	500 – 1500 Steps	0	Ok
IDU 1 SC	9^0	*0*	Ok
IDU 1 T PI		0	Ok
IDU 1 T PO		0	Ok
IDU 2 EEV	500 – 1500 Steps	0	Ok
IDU 2 SC	9^0	*0*	Ok
IDU 2 T PI		0	Ok
IDU 2 T PO		0	Ok
IDU 3 EEV	500 – 1500 Steps	0	Ok
IDU 3 SC	9^0	*0*	Ok
IDU 3 T PI		0	Ok
IDU 3 T PO		0	Ok
IDU 4 EEV	500 – 1500 Steps	0	Ok
IDU 4 SC	9^0	*0*	Ok
IDU 4 T PI		0	Ok
IDU 4 T PO		0	Ok

Directions:
Enter the data in the spaces provided. The reading will populate in the refrigerant diagram. Space temperature is entered at the IDU. Use the drop down boxes to identify the state of the component. Use the operating data and the target operation data on the next page to troubleshoot the system.

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DVM S 3 Phase Heat Recovery
Heating Operation Evaluation

Air Source Outdoor Unit				
Component	**Abbreviation**		**Heating Mode** Target	**Comments**
Compressor	Comp	Operation	High pressure control (default: 427 psig)	Primary function is to maintain head pressure
		Range	20~160Hz	
Outdoor Fan Motor	OFM	Operation	Low Pressure Control	Low Side saturation temperature is "typically" 2-10 degrees lower than outside air temperature
		Range	0-39 step (0-1300rpm)	
Main EEV	E_M	Operation	EVI Superheat control (default : SH 3.6°F -9°F)	Maintains evaorator superheat in the heating mode.
		Range	125 - 2000 Steps	
EVI EEV	E_EVI	Operation	On (Open)	
		Range	0-480step	
4 Way Valve	V_4W	Operation	On (Energized)	
EVI By-Pass Solenoid Valve	V_EB	Operation	On (Open): *T_a > 59°F or all compressors in unit are at 50Hz↓	Discharge temperature protection
			Off (Close): *T_a < 50°F and all compresors in unit are at 70Hz↑	Low Ambient heating Operation and Flash/ Vapor Injection Circuit controlled by unit
EVI Solenoid Valve	V_EVI	Operation	On (Open)	
Hot Gas By-Pass	V_HG	Operation	Off (Closed)	
Accumulator Oil Return Solenoid Valve	V_AR	Operation	On (Open when compressor is running)	**Off** during start up if DSH is < 18°F. **Off** for 2 min. if DSH is < 18°F **On** when DCSH > 27°F
Hot Gas By-Pass 2 (HR Only)	V_HG2	Operation	Off Closed	
Main EEV Solenoid Valve (HR Only) Main	V_ME	Operation	Off (Closed)	
Cooling Solenoid	VMC	Operation	Off (Closed)	
Condenser Out Thermistor	T_CO	Value	Should be 2 degrees lower than the outside air temperature	Typicaly it ranges between 2 to 10 degrees below outside air temperature

Mode Control Unit (MCU)				
Component	**Abbreviation**		**Heating Mode**	**Comments**
Port Heating Solenoid	Heat	Operation	On (Open)	On when in heating mode
Port Cooling Solenoid	Cool	Operation	Off (Closed)	
Port EEV	EEV	Operation	Closed 0 Steps	Open when a zone satisfies
		Range	0 - 480 Steps	
Sub-cooling EEV	SC_EEV	Operation	Closed 0 Steps	Active only in Heating Main and Cooling Main
		Range	480 Steps	
Sub-Cooler In Temperature		Operation		
Sub-Cooler Out Temperature		Operation	Close to Suction Header Temperature	

Indoor Unit IDU				
Component	**Abbreviation**		**Heating Mode**	**Comments**
Room Temperature	Room Temp.	Operation	61~86°F	
Liquid Pipe Thermistor	EVA In 1	Operation	Liquid Pipe Temperature for sub-cooling Control	Used to calculate Evaporator Sub-Cooling
Gas Pipe Thermistor	EVA Out 1	Operation	Hot Gas Pipe Temperature should be close to discharge temp.	Used to evaluate the temperature of the hot gas coming into the coil
EEV	EEV 1	Operation	Modulates to maintain 9 degrees of sub-cooling	Sub-Cooling = High Side Saturation Temperature - Pipe in Temp.
		Range	500- 1500 Steps (80 - 120 Steps when zone is satisfied)	Typical range between 500 and 1500 Steps
IDU Fan	Fan Speed	Range	Low , Medium, High, Auto	
		Operation	Low Speed: |Ts - Ti| ≤ 1.8°F	TS - Set Temperature
			Medium Speed: 1.8°F < |Ts - Ti| ≤ 5.4°F	Ti Indoor Temperature
			High Speed 5.4°F < |Ts - Ti|	